CUNEF UNIVERSIDAD

Econometrics 2024/2025

Introduction

Organizational issues

- E-mail: **andre.santos@cunef.edu**
- Office hours: Monday and Wednesday 11:00-13:00, either in person or via Zoom (send an email beforehand!)
- Software: The course will use R for all computer-based exercises.
- Evaluation:
  - Continuous evaluation 40%
    - 30% Paper-based midterm exam, in-class
    - 10% R-based group project
  - Final exam 60%
    - Covers all topics
    - Minimum grade is 5!

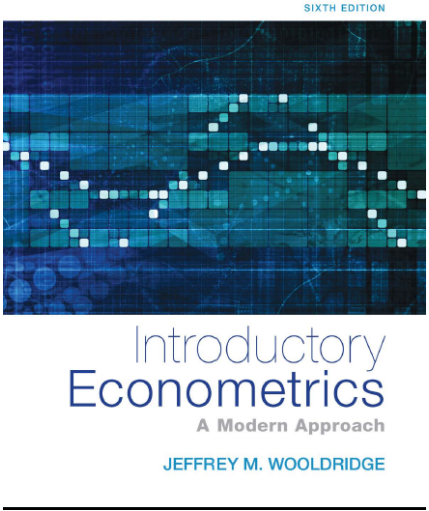
Organizational issues

- **Speaking in class**: You can only speak to contribute something to the class or to ask a question. People who disrupt the class will be expelled from the classroom.
- **Mobile phones**: They must remain in the backpack/bag during the class.
- **Laptops**: They can be used for taking notes. We will use the laptop for practical class activities.
- **Punctuality**: You cannot enter the classrooms once the class has started.

Organizational issues

EVERYTHING is in CANVAS

Bibliography



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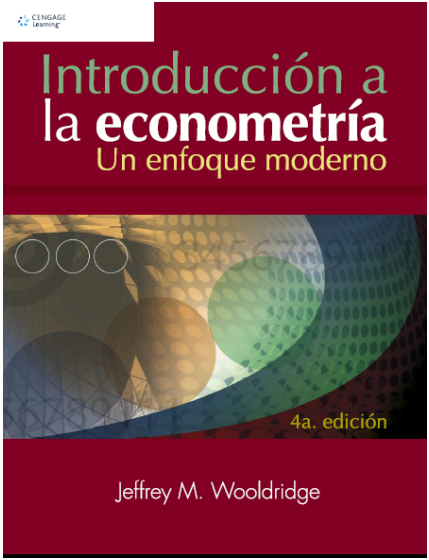
Bibliography



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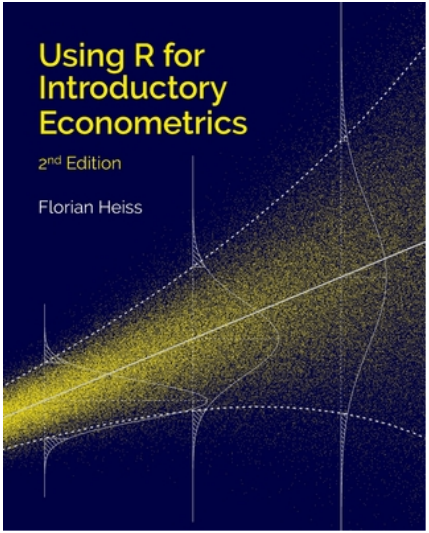
Bibliography



https://www.urfie.net/downloads/PDF/URfIE_web.pdf

What is Econometrics?

What is Econometrics?

- Econometrics is the use of statistical methods to analyze economic data (and other types of data).
- Econometrics may require a economic theory to be tested. It certainly requires **data**.
- Typical objectives of econometric analysis:
  - Estimation of relationships between economic variables.
    - **When inflation is high, is unemployment also high?**
  - Testing economic theories and hypotheses.
    - **Do high levels of inflation cause increases in unemployment?**
  - Evaluating and implementing government and business policies.
    - **How do Spanish firms react when the European Central Bank raises interest rates?**
  - Prediction / Forecasting.
    - **How high will unemployment and inflation be next year in Spain?**

Types of Data

- Cross-sectional data
- Time series data
- Panel or longitudinal data

Cross-sectional data sets

- These may include samples of individuals, households, firms, cities, states, countries, or other units of interest at a given point of time or in a given period.
- Example: Cross-sectional data set on wages and other characteristics

obsno	wage	educ	exper	female	married
1	3.10	11	2	1	0
2	3.24	12	22	1	1
3	3.00	11	2	0	0
4	6.00	8	44	0	1
5	5.30	12	7	0	1
.	.	.	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.
525	11.56	16	5	0	1
526	3.50	14	5	1	0

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Indicator variables (1 = yes, 0 = no)

Observation number

Hourly wage

Years of education

Years of experience

Time-series data sets

- This includes observations of a variable or several variables over time.
- Examples include stock prices, money supply, consumer price index, gross domestic product, annual homicide rates, automobile sales, and so on.
- Ordering of observations conveys important information.
- Data frequency may include daily, weekly, monthly, quarterly, annually, and so on.

	Date	Open	High	Low	Close	Daily Return	High-Low
FTSE-100 Index	6 July 2005	5190.10	5237.60	5190.10	5229.60	0.758%	47.50
	7 July 2005	5229.60	5229.60	5022.10	5158.30	–1.373%	207.50
	8 July 2005	5158.30	5232.20	5158.30	5232.20	1.422%	73.90

Panel or longitudinal data data sets

- The same cross-sectional units are followed over time.
- Panel data have a cross-sectional and a time series dimension.

	Date	Open	High	Low	Close	Daily Return	High-Low
FTSE-100 Index	6 July 2005	5190.10	5237.60	5190.10	5229.60	0.758%	47.50
	7 July 2005	5229.60	5229.60	5022.10	5158.30	–1.373%	207.50
	8 July 2005	5158.30	5232.20	5158.30	5232.20	1.422%	73.90
DAX-30 Index	6 July 2005	4607.57	4636.96	4607.57	4615.49	0.257%	29.39
	7 July 2005	4595.23	4595.23	4444.94	4530.18	–1.866%	150.29
	8 July 2005	4560.43	4597.97	4559.57	4597.97	1.485%	38.40
CAC-40 Index	6 July 2005	4272.64	4292.07	4264.00	4279.95	0.638%	28.07
	7 July 2005	4269.56	4269.77	4089.27	4220.62	–1.396%	180.50
	8 July 2005	4264.71	4300.31	4252.07	4300.31	1.871%	48.24

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